Bayesian Estimation of Historical Temperature Records in Seoul During Korean War

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Abstract

This project aims to estimate the missing monthly average temperature data for Seoul, South Korea, during the Korean War period (1950–1953). Due to the war, no data were recorded for nearly four years. We employ a probabilistic model (PyMC) to identify trends and seasonal effects, to generate posterior estimates for the four years period. The model is trained on average temperature data from 1908 to 2024. To validate model reliability, five additional four-year gaps were artificially introduced to re-fit the model for validation. Results indicate the model effectively recovers seasonal structure and long-term trends, although it under represents year to year variability. Despite this limitation, the model showed strong potential for recovering missing historical temperature records. Model could enhance accuracy by incorporating additional data and co-variate such as global trend.

1. Introduction

Each country maintains their historical climate records as they are essential for exploring climate trend and forecasting. Same applies to South Korea and its capital city, Seoul. South Korea has maintained an solid record of its meteorological data, including the temperature records of Seoul from 1907. This is even includes temperature data from Japanese occupation. However from 1950 to 1953, little less than 4 years data is missing due to tragic devastation caused by Korean War as no temperature data was recorded.

This short project aims to reconstruct the missing monthly temperature data for this 4 year span using a Bayesian modeling approach. We employ the probabilistic programming framework PyMC to generate statistically grounded estimates of monthly temperature data based on trend and seasonal effect from existing data while accounting for uncertainty.

Additionally, We will validate our result with existing data as well. We generate five other random consecutive four years period, remove them from training dataset, and generate artificial estimate in a same fashion. The predicted estimates of temperature for those 4 year span are then compared against the actual historical data.

2. Method

2-1. Data Source

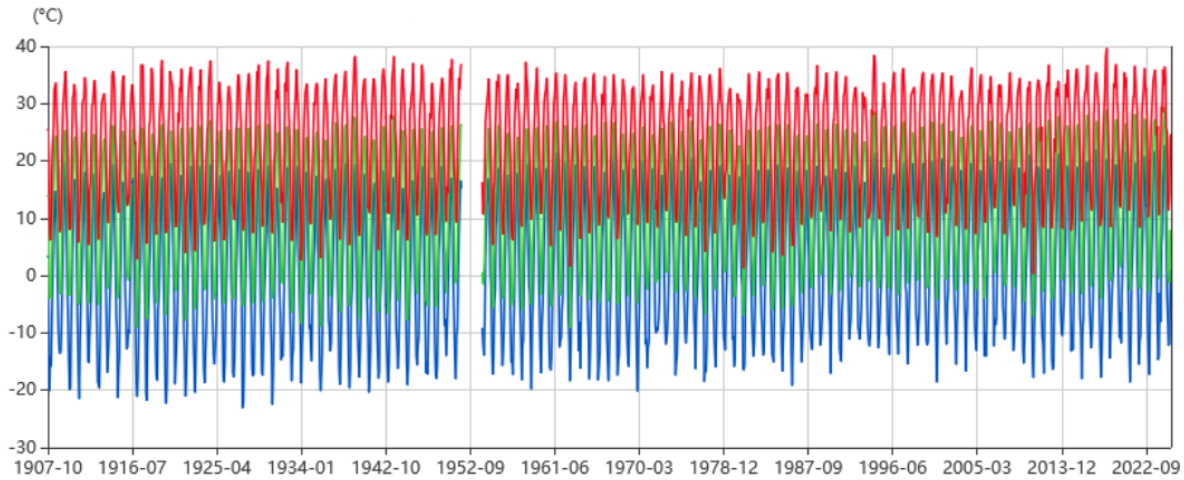

Figure 1: *Plot of average monthly temperature of the Seoul from 1907 to 2025. Red is average maximum temperature, green is average temperature, and blue is average minimum temperature of the each month. Generated and provided by data.kma.go.kr*

Data is collected from Korea Meteorological Administration data service (data.kma.go.kr). The dataset includes monthly maximum, minimum, and average temperature measurements for Seoul from 1908 to 2024. Only average temperature will be used in this project.

Prior to modeling, the data is standardized (zero mean and unit variance) to improve numerical stability during Bayesian inference. The missing period from 1950 to 1953 is masked and excluded from the modeling dataset.

2-2. Bayesian Model for Reconstruction

To estimate the missing temperature data in 4 year period, I used probabilistic Bayesian model that captures trend and seasonal effect. The observed standardized temperature y_t at time t is modeled as below equation:

$$y_t \sim \mathcal{N}(\mu_t + \gamma_{m(t)}, \sigma^2)$$

where:

- y_t: temperature (standardized) at time t.
- μ_t: latent trend at t.
- $\gamma_{m(t)}$: seasonal effect(monthly) for month $m(t) \in \{1, \ldots, 12\}$.
- σ: observation noise.

The latent trend μ_t is modeled using a Gaussian random walk,

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2_{\text{trend}})$$

Seasonal effects are modeled for each of the month. The constraint prevent seasonal effects are being biased from trend by centering seasonal effects.

$$\gamma_m \sim \mathcal{N}(0, \sigma^2_{\text{season}}), \quad \text{for } m = 1, \ldots, 12, \quad \sum_{m=1}^{12} \gamma_m = 0$$

Finally initial informative priors are roughly set using halfnormal distribution.

$$\sigma \sim \text{HalfNormal}(5.0)$$

$$\sigma_{\text{trend}} \sim \text{HalfNormal}(2.0)$$

$$\sigma_{\text{season}} \sim \text{HalfNormal}(2.0)$$

2-3. Verification of the model

Since we cannot directly assess the accuracy of the predicted dataset of 4 year period, we will have to utilize existing data to validate our approach. We randomly selected five 4 year period in 1908-2024, excluding already missing period from Korean War, and artificially removed those data.

Each of the five 4 year interval, we refit Bayesian model to predict posterior predictive estimates, the probabilistic guess for our artificial gap. Then estimates are compared to actual data using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). This validation approach will provides a quantitative measure of model accuracy.

3. Result and Discussion

3-1. Filling the missing gap

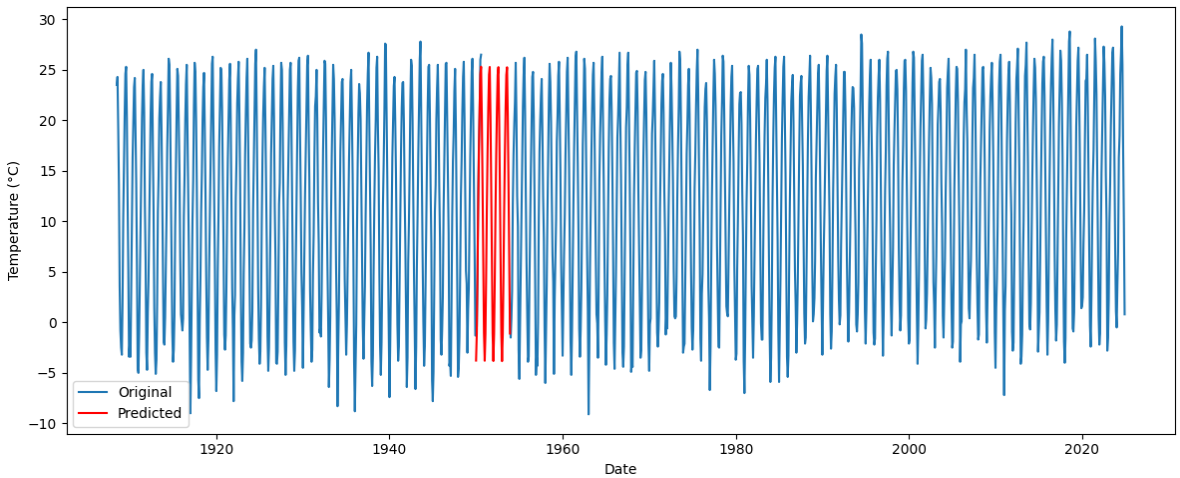

Figure 2: *Regenerated average temperature of missing period.*

Parameter	Mean	SD	HDI_2.5%	HDI_97.5%	MCSE_Mean	MCSE_SD	ESS_Bulk	ESS_Tail	$\hat{R}$
$\mu[0]$	-0.124	0.039	-0.201	-0.047	0.000	0.001	10719.421	6772.661	1.006
$\mu[1]$	-0.124	0.038	-0.199	-0.049	0.000	0.001	10854.858	7033.867	1.006
$\mu[2]$	-0.123	0.036	-0.198	-0.055	0.000	0.001	9990.371	6898.552	1.004
$\mu[3]$	-0.123	0.035	-0.194	-0.057	0.000	0.001	10107.265	6915.453	1.003
$\mu[4]$	-0.124	0.034	-0.194	-0.061	0.000	0.000	10722.729	6460.570	1.003
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\gamma[10]$	-0.516	0.331	-1.188	0.138	0.003	0.005	16125.871	5686.991	1.001
$\gamma[11]$	-1.236	0.331	-1.905	-0.578	0.003	0.005	16355.546	5722.387	1.001
σ	0.132	0.003	0.126	0.138	0.000	0.000	46.172	407.933	1.061
σ_{trend}	0.013	0.004	0.007	0.019	0.001	0.000	10.618	42.721	1.336
σ_{season}	1.136	0.274	0.684	1.677	0.004	0.005	7097.467	4774.380	1.001

Table 1: Summary statistics for posterior parameters

Using the Bayesian model we generated missing posterior distributions for the monthly average temperatures of Seoul during the Korean War period (1950–1953). Generated posterior distributions looked smooth and maintain seasonal pattern that aligned with adjacent years. It captured clear patterns of colder winter months and warmer summer months, and estimated distribution seems to capture long-term trend very well. Rhat values were close to 1 signifying model has converged.

However model also followed by some significant limitations. Although model effectively capture long-term trend and seasonal pattern of winter and summer, at the same time it failed to capture year to year variance or fluctuation. As you can see in the figure 2, some years exerted unexpectedly high or low temperatures. similar was reported in 1951, where South Korean and UN soldiers have reported severe non-combat loss due to harsh winter that year. Yet our failed to capture that aspect as it seems to evaluate trend more higher

and underestimate year to year fluctuations despite trend constant being very low.

This suggests our model still has room for improvement. Our model could benefit by incorporating additional co-variate, such as average temperature fluctuation from different cities in South Korea such as Busan.

3-2. Validation

To validate the model, we re-fit the model with additional 4 year mask on the dataset as described in the method section. Five random 4 year intervals were selected, masked, and estimated. The reconstructed temperatures were compared against the actual historical temperature data.

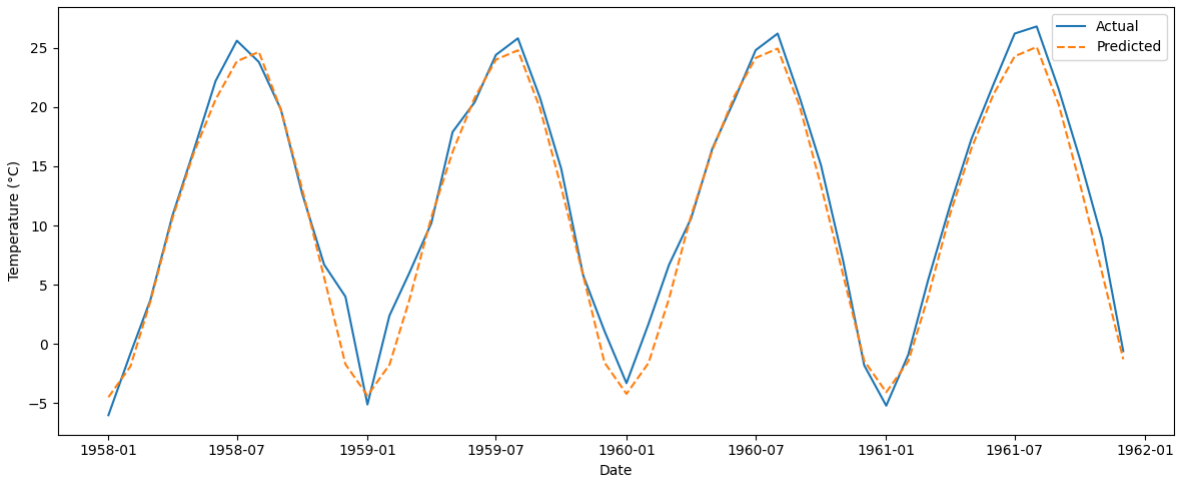

Figure 3: *Comparison between predicted vs. actual temperature data during 1958-1961*

Window	**RMSE**	**MAE**
1958–1961	1.679081	1.262648
1963–1966	1.400474	0.984367
1931–1934	1.500569	1.141393
1974–1977	1.426963	1.109786
2007–2010	1.316538	1.015012
Average	**1.464725**	**1.102641**

Table 2: RMSE and MAE for various windows

Validation result has confirmed that model is a reasonable tool for estimating missing temperature in 4 year period. As one example for year 1958-1961 shows in Figure 3, estimates were almost perfect in some cases. In general, the Root Mean Squared Error (RMSE) remained around 1.46, and the Mean Absolute Error (MAE) maintained similar patterns.

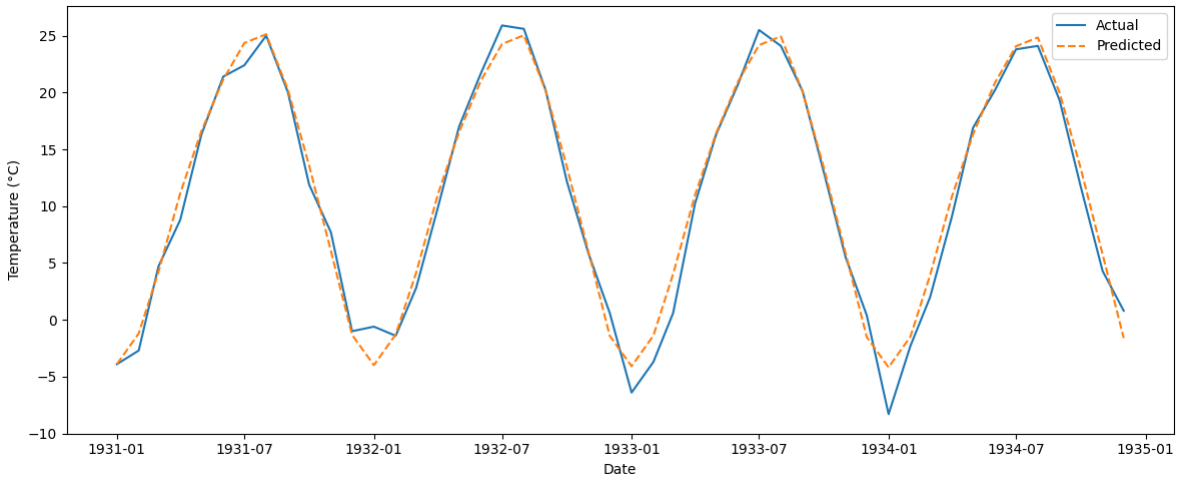

Figure 4: *Comparison between predicted vs. actual temperature data during 1931-1934*

Yet validation showed limitation as well, as model lacks in year specific variance, it cannot represent exceptionally harsh summer or winter in a specific year For figure 4, we can see that for year 34, estimate for January is off by almost 5 degree Celsius. Model's estimate relies on seasonal pattern and general long-term trend thus it tends to over-smooth curves in certain years, making year specific fluctuation even harder to estimate in current model. In general our model was more accurate in estimating summer seasonal pattern over winter as well.

In conclusion, validation has confirmed reliability of the model, but also shown limitation of this model as well. Although model currently relies more heavily on seasonal constant over trend, it still overfit on trend as it fail to represent year specific variance. Incorporating additional external data and co-variate, such as global or regional maximum and minimum temperature of each year, into the model could improve the prediction by introducing year specific variance.

4. Conclusion

This project reconstructed missing temperature data for Seoul, Korea during Korean War period using posterior estimate of Bayesian probabilistic model. By modeling long-term trends and seasonal patterns model was able to re-generate rough estimate of the temperature of each month during this period. Validation using existing dataset suggested that model has successfully captured seasonal pattern in its estimate.

Although model performed well enough to provide rough estimate, it still has tenancy to over-represent trend and ignore year to year fluctuation. Incorporating additional data such as annual temperature average of global or near by city into the model could improve this model to reflect more accurate climate behaviors.

War is a tragedy that results only in destruction and loss. In this effort to revive the climate data from that time of suffering, I hope to demonstrate the resilience of humanity and the history—especially for those facing desperate times today.